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A Simple Transformation Approach to Difference-in-Differences Estimation for Panel Data

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ABSTRACT

In the case of panel data with staggered interventions, we propose a simple time-series transformation that can be combined with various treatment effect estimators. The transformation is at the unit level, and simply requires computing the average outcome prior to an intervention, subtracting it from a post-treatment outcome, and then carefully selecting the control units in each time period. We show that under no anticipation and parallel trends assumptions, the cohort treatment indicators satisfy the key unconfoundedness assumption with respect to the transformed potential outcome. Given identification, any number of treatment effect estimators, including matching estimators, can be applied for each treated cohort and calendar time pair where the average treatment effects on the treated are identified. The doubly robust method of combining inverse probability weighting with linear regression adjustment works particularly well in terms of bias and efficiency. Importantly, our transformation is easily modified to account for unit-specific trends, allowing relaxation of the parallel trends assumption even after conditioning on covariates.

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Difference-in-differences; Doubly robust estimators; Heterogeneous trends; Panel data; Parallel trends

1. Introduction

The two-way fixed effects (TWFE) estimator, applied to a linear panel model with a constant treatment effect, has been regularly applied in difference-in-differences settings. The estimator is easy to understand and is taught in courses that cover panel data methods. Recently, several authors have pointed out shortcomings of the constant effect model under staggered interventions, including Borusyak, Jaravel, and Spiess (2024), Goodman-Bacon (2021), and de Chaisemartin and D'Haultfœuille (2020). These papers derive alternative representations of the simple TWFE estimator when the treatment effects are heterogeneous across treatment cohorts or time periods.

Other authors have proposed more flexible estimation methods that uncover average treatment effects on the treated in the staggered intervention case. These include Callaway and Sant'Anna (2021), who propose long-differencing strategies and apply standard treatment effect estimators. Sun and Abraham (2021) propose a fixed effects estimator applied to a more flexible model. Both Sun and Abraham (2021) and Callaway and Sant'Anna (2021) are event-study, or "leads and lags" estimators that use only the single period prior to the first intervention time as the reference period. As recently discussed in Wooldridge (2025b), when the Sun and Abraham (2021) approach is extended to allow covariates to appear in a fully flexible way, the resulting estimator is the same as the Callaway and Sant'Anna (2021) pairwise regression adjustment estimators using the never-treated group as the control group. The label "leads and lags" comes from the fact that the regression includes the lead values of the treatment indicators in addition to the lags. Wooldridge (2025b) contrasts the leads and lags approach with the "lags only" approach, which omits all pre-treatment indicators and, in effect, averages all pre-treatment periods to construct a reference period. In both cases, one includes a full set of interactions between the covariates and any of the dummy variables included in the regression. Wooldridge (2025b) also shows the pooled ordinary least squares (POLS) estimator is equivalent to the TWFE estimator on the same equation, but where the time-constant cohort dummies and time dummies are swept away by the TWFE transformation. Borusyak, Jaravel, and Spiess (2024) propose imputation estimators based on pooled OLS regressions, where the first includes unit and time fixed effects. Wooldridge (2025b) shows that, with time-constant covariates and a balanced panel, the imputation estimates are identical to the POLS (and therefore TWFE) estimates in the flexible model using the entire sample.

An attractive feature of the Callaway and Sant'Anna (2021) approach, which builds on Abadie (2005) and Sant'Anna and Zhao (2020) for the two-period case, is that it permits the application of treatment effects estimators beyond regression adjustment. However, as mentioned above, the Callaway and Sant'Anna (2021) method uses only the period just prior to the intervention in defining a reference period, thereby discarding potentially useful information in earlier time periods. Wooldridge (2025b) shows that, under the standard "error components" structure, with a homoskedastic time-constant component and homoskedastic and serially uncorrelated idiosyncratic

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errors, the lags only POLS estimator is both best linear unbiased (BLUE) and asymptotically efficient. These results imply that the Callaway and Sant'Anna (2021) estimators are less efficient under a standard set of assumptions. On the other hand, under strong forms of positive serial correlation, Callaway and Sant'Anna (2021) can be more efficient because it is similar to a first-differencing estimator. See Harmon (2024), de Chaisemartin and d'Haultfoeuille (2023), and Wooldridge (2025b) for further discussion.

In this paper, we propose an alternative "rolling" approach that uses information on all pre-treatment outcomes while permitting application of a variety of treatment effects estimators, including regression adjustment (RA), inverse probability weighting (IPW), and the doubly robust IPWRA proposed by Wooldridge (2007). If one prefers, matching on covariates or the propensity score can be used instead. Just as the Callaway and Sant'Anna (2021) can be viewed as applying treatment effects estimators in place of regression in the leads and lags case, our approach can be viewed as extending the lags only regression methods to various treatment effect estimators.

Another contribution of this paper is that we show how a slight modification of our identification argument establishes identification when (long) differencing is used, as in Callaway and Sant'Anna (2021). Our identification argument in Section 4 is not tied to a particular estimation method, and so we extend the Callaway and Sant'Anna (2021) class of estimators to include IPWRA estimators, matching estimators, and even causal machine learning estimators. This extension allows one to make direct comparisons between our rolling approach, which uses all pre-treatment periods, and the Callaway and Sant'Anna (2021) approach, which uses only the single period before the intervention. We view the approaches as complementary in much the same way that the lags only and leads and lags regression-based estimators described in Wooldridge (2025b) are complementary. Both approaches yield consistent estimators under no anticipation and a popular (conditional) parallel trends (PT) assumption, and they generally have different biases when PT fails. Moreover, neither estimator is guaranteed to be more efficient than the other across empirically relevant settings. Because we show that the same estimation methods, including doubly robust estimators, can be applied to either transformation, it makes sense to try both in empirical research.

In the case of common timing, we show that applying regression adjustment to our transformed outcome variable is equivalent to the regression adjustment estimator based on levels. As mentioned above, Wooldridge (2025b) shows that this estimator is both BLUE and asymptotically efficient under standard assumptions. This provides motivation for applying estimators other than regression adjustment to the transformed variables in order to check robustness of findings. In the case of staggered entry, our approach identifies the average treatment effects on the treated (ATTs) by cohort and calendar time under the same no anticipation and parallel trends assumptions in Wooldridge (2025b). Once identification is established, various estimation methods can be applied.

The remainder of the paper is organized as follows. Section 2 begins with the common timing case, defining the potential outcomes and parameters of interest, and establishing identification assumptions. We start with this case for both pedagogical and theoretical reasons. It is easiest to see the workings of our approach, and to derive certain equivalences, in the common timing case. Moreover, the common timing setting allows straightforward efficiency comparisons with competing approaches. In Section 3 we propose a general approach to estimation using a transformed outcome variable. In Section 4 we extend the framework and identification argument to the staggered case. Section 4.4 discusses how one might accommodate suspected failures of no anticipation, and how one modifies the procedure for unbalanced panels. In Section 5, we show how we can account for heterogeneous trends, focusing on linear trends, to allow violation of the parallel trends assumption (even after we condition on covariates)—a possibility not allowed in Callaway and Sant'Anna (2021). In Section 6, we revisit the literature on the effects of Walmart openings on local labor markets. Section 7 contains some concluding remarks. Appendix C of the Online Appendix reports Monte Carlo simulations showing that the rolling methods perform well in terms of both bias and precision.

2. Setup and Identification with Common Timing

We begin with the case where the date of the intervention is the same for all treated units, and the intervention is in place through the final period. The time periods in the population are $t = 1, 2, \ldots, T$ and the date of the intervention is S, where $1 < S \leq T$; we assume at least one pre-treatment period. The identification arguments are based on an underlying population, and for each unit i, we use $\{Y_{it}(d) : d \in \{0, 1\},\ t = 1, \ldots, T\}$ to denote the time series of potential outcomes, where $d = 0$ denotes the control state and $d = 1$ denotes the treated state. Depending on the context, we suppress the unit index i and write $Y_t(d)$ in place of $Y_{it}(d)$ for notational simplicity.

The binary time-constant treatment indicator is D, where $D = 1$ means treatment starting in period S and lasting through period T. A time-varying treatment indicator is $W_t = D \cdot p_t$, where p_t is a post-treatment indicator equal to 1 if $t \geq S$ and otherwise zero.

Like the recent literature on DiD, our focus is on the average treatment effect on the treated (ATT or ATET) in each treated period $r = S, \ldots, T$:

$$\tau_r = E\left[Y_r(1) - Y_r(0) \mid D = 1\right]. \tag{2.1}$$

The fundamental problem of identification of τ_r is that we only observe the treatment status, D, and the outcome,

$$Y_r = (1 - D) \cdot Y_r(0) + D \cdot Y_r(1). \tag{2.2}$$

Importantly, when $D = 1$, $Y_r = Y_r(1)$, which means

$$E\left[Y_r(1) \mid D = 1\right] = E\left(Y_r \mid D = 1\right), \quad r = S, \ldots, T. \tag{2.3}$$

The expectation $E(Y_r \mid D = 1)$ can be estimated in a consistent, even unbiased, way under various sampling schemes. Under random sampling, the average of Y_r across the treated subsample is unbiased and consistent. Therefore, writing

$$\begin{aligned}\tau_r &= E\left[Y_r(1) \mid D = 1\right] - E\left[Y_r(0) \mid D = 1\right] \\ &= E\left(Y_r \mid D = 1\right) - E\left[Y_r(0) \mid D = 1\right],\end{aligned}$$

it is easily seen that the challenge is in identifying $E[Y_r(0) \mid D = 1]$.

If the treatment is randomly assigned then estimation of $E[Y_r(0) \mid D = 1]$ is easy because, under randomization,

$$E[Y_r(0) \mid D = 1] = E[Y_r(0) \mid D = 0] = E(Y_r \mid D = 0).$$

For most applications, random assignment is too strong an assumption. To see how to relax it, use simple algebra to write

$$\begin{aligned} Y_r(1) - Y_r(0) &= \left[Y_r(1) - \frac{1}{(S-1)} \sum_{q=1}^{S-1} Y_q(1) \right] \\ &\quad - \left[Y_r(0) - \frac{1}{(S-1)} \sum_{q=1}^{S-1} Y_q(0) \right] \\ &\quad + \frac{1}{(S-1)} \sum_{q=1}^{S-1} \left[Y_q(1) - Y_q(0) \right] \\ &\equiv \dot{Y}_r(1) - \dot{Y}_r(0) + \frac{1}{(S-1)} \sum_{q=1}^{S-1} \left[Y_q(1) - Y_q(0) \right] \end{aligned} \tag{2.4}$$

where

$$\dot{Y}_r(1) \equiv Y_r(1) - \frac{1}{(S-1)} \sum_{q=1}^{S-1} Y_q(1) \tag{2.5}$$

and similarly for $\dot{Y}_r(0)$. Note that for each $r \in \{S, S+1, \ldots, T\}$, $\dot{Y}_r(1)$ denotes the treated potential outcome in period r, after subtracting the average of the pre-treatment outcomes. The third term is the average of the difference of the pre-treatment period "treatment" effects.

Given the representation in (2.4), we can write

$$\begin{aligned} \tau_r &= E\left[\dot{Y}_r(1) \mid D = 1\right] - E\left[\dot{Y}_r(0) \mid D = 1\right] \\ &\quad + \frac{1}{(S-1)} \sum_{q=1}^{S-1} E\left[Y_q(1) - Y_q(0) \mid D = 1\right] \end{aligned} \tag{2.6}$$

The first assumption, a "no anticipation" condition, eliminates the third term in (2.6).

Assumption 2.1. **NAC (No Anticipation, Common Timing).** For the eventually treated indicator D,

$$E[Y_t(1) - Y_t(0) \mid D = 1] = 0, \quad t = 1, \ldots, S-1. \tag{2.7}$$

The name of this assumption derives from the fact that $E[Y_t(1) - Y_t(0) \mid D = 1]$ for $t < S$ are average treatment effects on the treated prior to the intervention, and the assumption is that these are all zero. Assumption 2.1 is implied by an assumption commonly used in the literature, namely, $Y_t(1) = Y_t(0)$, $t = 1, \ldots, S-1$. This assumption is implicit in Heckman, Ichimura, and Todd (1997) and made explicit in Abadie (2005) and elsewhere. Because the variable indexing $Y_t(\cdot)$ is treatment status not yet assigned, the assumption rules out anticipatory changes in the potential outcomes, on average. If one is concerned that the announcement of a policy prior to its implementation may result in some units strategically manipulating their pre-intervention outcomes, one might drop a period or two just prior to the intervention.

Given Assumption 2.1, we can express τ_r as

$$\tau_r = E\left[\dot{Y}_r(1) \mid D = 1\right] - E\left[\dot{Y}_r(0) \mid D = 1\right]. \tag{2.8}$$

Estimating the first term in (2.8) is easy because we observe $\dot{Y}_r(1)$ when $D = 1$. More precisely, define the same transformation in the observed variable Y_r:

$$\dot{Y}_r \equiv Y_r - \frac{1}{S-1} \sum_{q=1}^{S-1} Y_q \equiv Y_r - \overline{Y}_{pre} \tag{2.9}$$

When $D = 1$, $\dot{Y}_r = \dot{Y}_r(1)$, and so $E[\dot{Y}_r(1) \mid D = 1] = E(\dot{Y}_r \mid D = 1)$. The latter is trivially identified (as usual, under a suitable sampling scheme). Notice that, in the simple $T = 2$ case with $S = 2$, $\dot{Y}_2 = Y_2 - Y_1$, which is the difference from period one to period two.

The difficult term in identifying τ_r is $E[\dot{Y}_r(0) \mid D = 1]$. The unconditional parallel trends assumption implies that $E[\dot{Y}_r(0) \mid D = 1] = E[\dot{Y}_r(0) \mid D = 0]$. Here we follow the recent literature, for example Callaway and Sant'Anna (2021), and assume that parallel trends holds after conditioning on observed pre-treatment covariates.

Assumption 2.2. **CPTC (Conditional Parallel Trends, Common Timing).** For observed covariates $\mathbf{X}$,

$$\begin{aligned} &E[Y_t(0) - Y_1(0) \mid D, \mathbf{X}] = E[Y_t(0) - Y_1(0) \mid \mathbf{X}], \\ &\quad t = 2, \ldots, T. \end{aligned} \tag{2.10}$$

Conceptually, we can think of the CPTC assumption as meaning that parallel trends holds within sub-partitions of the population as defined by $\mathbf{X}$. We do not require that the PT assumption hold across the entire population.

Simple algebra shows that (2.10) is the same as assuming $E[Y_t(0) - Y_s(0) \mid D, \mathbf{X}] = E[Y_t(0) - Y_s(0) \mid \mathbf{X}]$ for all $t \neq s$. Wooldridge (2025b) used very similar assumptions, along with linearity of conditional means, to derive identification of the τ_r. Here, we are interested in applying methods other than linear regression adjustment to the transformed outcomes in (2.9). Assumption 2.2 allows us to identify $E[\dot{Y}_r(0) \mid D = 1]$. To see how, first note that, by iterated expectations,

$$E\left[\dot{Y}_r(0) \mid D = 1\right] = E\left\{E\left[\dot{Y}_r(0) \mid D = 1, \mathbf{X}\right] \mid D = 1\right\} \tag{2.11}$$

Next, write

$$\dot{Y}_r(0) = (S-1)^{-1} \sum_{q=1}^{S-1} \left[Y_r(0) - Y_q(0)\right].$$

Then, by CPTC,

$$\begin{aligned} &E\left[\dot{Y}_r(0) \mid D = 1, \mathbf{X}\right] \\ &\quad = (S-1)^{-1} \sum_{q=1}^{S-1} E\left[Y_r(0) - Y_q(0) \mid D = 1, \mathbf{X}\right] \\ &\quad = (S-1)^{-1} \sum_{q=1}^{S-1} E\left[Y_r(0) - Y_q(0) \mid D = 0, \mathbf{X}\right] \\ &\quad = E\left[Y_r(0) - (S-1)^{-1} \sum_{q=1}^{S-1} Y_q(0) \,\middle|\, D = 0, \mathbf{X} \right] \\ &\quad = E\left[\dot{Y}_r(0) \mid D = 0, \mathbf{X}\right]. \end{aligned} \tag{2.12}$$

The conclusion in (2.12) is simple but important. It says that, in terms of the potential outcome $\dot{Y}_r(0)$, treatment D is unconfounded conditional on $\mathbf{X}$. Assumption 2.1 ensures that the period-specific ATTs can be expressed as in (2.8). This means that, for a post-intervention period r, we have turned the difference-in-differences problem into a standard problem of estimating an ATT in a cross-sectional population.

Using the fact that $Y_q = Y_q(0)$ when $D = 0$, (2.12) implies that,

$$E[\dot{Y}_r(0) \mid D = 1, \mathbf{X}] = E(\dot{Y}_r \mid D = 0, \mathbf{X}).$$

Now the argument is the same as in a standard cross-sectional treatment effects setting:

By iterated expectations,

$$\begin{aligned} E[\dot{Y}_r(0) \mid D = 1] &= E[E(\dot{Y}_r \mid D = 0, \mathbf{X}) \mid D = 1] \\ &\equiv E[\dot{m}_{0r}(\mathbf{X}) \mid D = 1], \end{aligned} \tag{2.13}$$

where $\dot{m}_{0r}(\mathbf{X}) \equiv E(\dot{Y}_r \mid D = 0, \mathbf{X} = \mathbf{x})$ is the conditional mean of the observed variable $\dot{Y}_r$ for the control group. This function is nonparametrically identified on Supp $(\mathbf{X} \mid D = 0)$, the support of the covariates for the control group. To ensure that $E[\dot{m}_{0r}(\mathbf{X}) \mid D = 1]$ can be computed without extrapolating to covariate values outside Supp$(\mathbf{X} \mid D = 0)$, we impose a standard overlap assumption.

Assumption 2.3. **OVLC (Overlap, Common Timing).** Define the propensity score

$$p(\mathbf{x}) = P(D = 1 \mid \mathbf{X} = \mathbf{x}), \mathbf{x} \in \text{Supp}(\mathbf{X}). \tag{2.14}$$

Then

$$p(\mathbf{x}) < 1, \mathbf{x} \in \text{Supp}(\mathbf{X}). \tag{2.15}$$

The previous derivations and discussion prove the following.

Theorem 2.1. Under Assumption 2.1, τ_r can be expressed as in (2.8) for $r = S, \ldots, T$. Under Assumption CPTC, D is unconfounded (in the conditional mean sense) with respect to $\dot{Y}_r(0)$ conditional on $\mathbf{X}$. When we add Assumption 4.3, the parameters $\tau_r, r = S, \ldots, T$, are identified.

3. Estimation in the Common Timing Case

Given the identification result stated in Theorem 2.1, estimation of the τ_r is straightforward. Once the outcome is transformed as in (2.9), we can apply any estimation method. Essentially, this is the conclusion reached in Sant'Anna and Zhao (2020) in the $T = 2$ case. Earlier, Abadie (2005) proposed inverse probability weighting when $T = 2$.

Assume we observe a random sample of size N from the cross section. The observed outcome can be expressed as

$$Y_{it} = (1 - D_i) \cdot Y_{it}(0) + D_i \cdot Y_{it}(1), \tag{3.1}$$

where the i subscript denotes unit i. For each i, we observe $\{(Y_{it}, D_i, \mathbf{X}_i) : i = 1, 2, \ldots, N\}$. To exploit the unconfoundedness result in Theorem 2.1, we simply need to obtain the transformed data. For each unit i, define

$$\dot{Y}_{ir} = Y_{ir} - \frac{1}{(S-1)} \sum_{q=1}^{S-1} Y_{iq} \equiv Y_{ir} - \overline{Y}_{i,pre}. \tag{3.2}$$

Then, for any $r \in \{S, \ldots, T\}$, we can apply any standard treatment effect (TE) estimator to the data

$$\{(\dot{Y}_{ir}, D_i, \mathbf{X}_i) : i = 1, 2, \ldots, N\}.$$

A common TE estimator is (linear) "regression adjustment" (RA), which means estimating separate regression functions for the control and treated units. While the RA estimator of τ_r can be characterized using imputation, as shown in Wooldridge (2025b), it can instead be obtained from a single regression using all observations in period r:

$$\dot{Y}_{ir} \text{ on } 1, D_i, \mathbf{X}_i, D_i \cdot \left(\mathbf{X}_i - \overline{\mathbf{X}}_1\right), \ \ i = 1, 2, \ldots, N, \tag{3.3}$$

where $\hat{\tau}_r$ is the coefficient on D_i and $\overline{\mathbf{X}}_1 = N_1^{-1} \sum_{i=1}^{N} D_i \mathbf{X}_i$ is the vector of averages over the treated units. Obtaining $\hat{\tau}_r$ from a single regression means that one can use standard software once the $\dot{Y}_{ir}$ have been obtained. This leads to simple inference for $\hat{\tau}_r$, allowing easy computation of standard errors robust to any kind of heteroscedasticity. Also, it is often easy to account for the sampling variation in $\overline{\mathbf{X}}_1$ as an estimator of $\mu_1 \equiv E(\mathbf{X} \mid D = 1)$. Issues of clustering standard errors are relatively easy to deal with given we have a standard cross-sectional regression.

Because of the representation of $\dot{Y}_{ir}$ in (3.2), there is a simple characterization of $\hat{\tau}_r$. All of the coefficients in (3.3) are obtained by differencing the coefficients from two separate regressions. In the first, Y_{ir} is regressed on all variables in (3.3). Then $\overline{Y}_{i,pre}$ is regressed on the same set of variables, and these coefficients are subtracted from the first. In particular, $\hat{\tau}_r$ is obtained as the difference between two standard ATT estimators using regression adjustment. The first uses observations in period r only, and the second uses the average of Y_{iq} over the pre-treatment periods. Without covariates, the estimator would be

$$\hat{\tau}_r = \left(\overline{Y}_{1r} - \overline{Y}_{0r}\right) - \left(\overline{Y}_{1,pre} - \overline{Y}_{0,pre}\right), \tag{3.4}$$

where the first subscript indicates treated or control status. This has a clear interpretation as a DiD estimator.

Recognizing that an estimator can be obtained from (3.3) has additional benefits. For example, if D_i is independent of $\dot{Y}_{ir}(0)$—so that the unconditional PT assumption holds—then the covariates $\mathbf{X}_i$ need not be included in (3.3) for the coefficient on D_i to consistently estimate τ_r. If, in addition, D_i is independent of $\mathbf{X}_i$, the regression in (3.3) still can be used to improve efficiency over the simple estimator without the covariates. As discussed in Negi and Wooldridge (2021), such improvements are possible if $\mathbf{X}_i$ helps predict $\dot{Y}_{ir}$. In many cases, $\mathbf{X}_i$ may not have much predictive power for $\dot{Y}_{ir}$ even though it might predict the level, Y_{ir}, well.

It turns out there is another useful algebraic equivalence. Suppose we act *as if* the following conditional expectation holds for all population units and time periods:

$$\begin{aligned} E(Y_t \mid D, \mathbf{X}) = {} & \alpha + \mathbf{X}\beta + \gamma D + (D \cdot \mathbf{X})\delta + \sum_{r=2}^{T} \theta_r fr_t \\ & + \sum_{r=2}^{T} \left(fr_t \cdot \mathbf{X}\right)\pi_r + \sum_{r=S}^{T} \tau_r \left(D \cdot fr_t\right) \\ & + \sum_{r=S}^{T} \left(D \cdot fr_t\right)(\mathbf{X} - \mu_1)\eta_r, \ \ t = 1, \ldots, T, \end{aligned} \tag{3.5}$$

where fr_t is a time period dummy equal to one if $r = t$ and zero otherwise. The interaction $D \cdot fr_t$ is the treatment indicator for time period r. Equation (3.3) suggests a pooled OLS regression across all i and t:

$$\begin{aligned}&Y_{it} \text{ on } 1, \mathbf{X}_i, D_i, D_i \cdot \mathbf{X}_i, f2_t, \ldots, fT_t, f2_t \cdot \mathbf{X}_i, \ldots, fT_t \cdot \mathbf{X}_i\\ &D_i \cdot fS_t, \ldots, D_i \cdot fT_t, D_i \cdot fS_t \cdot \left(\mathbf{X}_i - \overline{\mathbf{X}}_1\right), \ldots,\\ &D_i \cdot fT_t \cdot \left(\mathbf{X}_i - \overline{\mathbf{X}}_1\right). \qquad (3.6)\end{aligned}$$

The estimated treatment effects, say $\tilde{\tau}_r$, are the coefficients on the treatment dummies $D_i \cdot fS_t, \ldots, D_i \cdot fT_t$. Wooldridge (2025b) shows that the $\tilde{\tau}_r$ are numerically identical to a two-stage imputation approach based on the levels, Y_{it}. It turns out that the $\tilde{\tau}_r$ are also equivalent to the $\hat{\tau}_r$ obtained by using the transformed outcome variable, $\dot{Y}_{ir}$, one period at a time.

Theorem 3.1. Let $\hat{\tau}_r$, $r = S, S+1, \ldots, T$ be the coefficients on D_i in the separate regressions (3.3) and let $\tilde{\tau}_r$ be the coefficients on $D_i \cdot fS_t, \ldots, D_i \cdot fT_t$ from (3.6). Then $\tilde{\tau}_r = \hat{\tau}_r, r = S, S+1, \ldots, T$. Moreover, the coefficient vector on $D_i \cdot fr_t \cdot \left(\mathbf{X}_i - \overline{\mathbf{X}}_1\right)$ in (3.6) is identical to that on $D_i \cdot \left(\mathbf{X}_i - \overline{\mathbf{X}}_1\right)$ in (3.3).

The proof of Theorem 3.1 can be found in Online Appendix A. The equivalence is valuable for a couple of reasons. First, it shows that two different ways to approach identification under the same set of assumptions—that in Wooldridge (2025b) and the approach we use here—lead to the same estimation methods. Second, Wooldridge (2025b, Theorem 6.2) shows that, under standard assumptions on the implied error term, which includes a unit-specific unobserved effect and a time-varying component, the estimators from (3.6) are both best linear unbiased and asymptotically efficient with T fixed and $N \to \infty$, under random sampling across i. This establishes that the transformation used in (3.3) does not discard useful information.

Given the equivalence of our transformation approach and the OLS estimator pooled across i and t, what use is the former? Importantly, it allows us to use other treatment effects estimators beyond regression adjustment. For example, we can apply IPW or IPWRA using the cross-sectional data $\{(\dot{Y}_{ir}, D_i, \mathbf{X}_i) : i = 1, 2, \ldots, N\}$. We can also apply propensity score matching or covariate matching.

Procedure 3.1 **(Rolling Methods, Common Timing).** Step 1. For a given time period $r \in \{S, \ldots, T\}$ and each unit i, compute $\dot{Y}_{ir}$ as in (3.2).

Step 2. Using all of the units, apply standard TE methods—such as linear RA, IPW, IPWRA, matching—to the cross section

$$\left\{\left(\dot{Y}_{ir}, D_i, \mathbf{X}_i\right) : i = 1, \ldots, N\right\}.$$

Inference on a single cohort-time treatment effect is straightforward in Step 2 using built-in Stata commands such as `teffects`. The method of obtaining the IPWRA estimator using a weighted least squares regression, with IPW as a special case, is described in Section 19.4 of Wooldridge (2025a). After the rolling transformation, the IPWRA estimator can be viewed as a two-step M-estimator. Under standard regularity conditions for two-step M-estimation (e.g., Wooldridge 2007; Newey and McFadden 1994)—including overlap, finite second moments of the relevant estimating equations (e.g., $E[Y^2] < \infty$ and $E[\|X\|^2] < \infty$), nonsingularity of the associated moment matrices, and sufficient regularity of the first-step estimators—consistency and asymptotic normality follow. For IPWRA, consistency requires correct specification of either the propensity score model or the outcome model.

We can compare the transformation in (3.2) to that in Callaway and Sant'Anna (2021). In the common timing case, the CS transformation, for $r \geq S$, is

$$\mathring{Y}_{ir} = Y_{ir} - Y_{i,S-1}, \qquad (3.7)$$

so that $\mathring{Y}_{ir}$ is a "long" difference. When $T = 2$, the transformation is $\dot{Y}_2 = \mathring{Y}_2 = \Delta Y_2 \equiv Y_2 - Y_1$. Thus, our approach encompasses and extends Abadie (2005) and Sant'Anna and Zhao (2020) in the panel data case.

4. Staggered Interventions

4.1. Some Units Never Treated

We now turn to the staggered intervention case. As in Athey and Imbens (2022) and Wooldridge (2023, 2025b), the potential outcomes are denoted

$$Y_t\left(g\right), g \in \{S, \ldots, T, \infty\}, \quad t \in \{1, 2, \ldots, T\}, \qquad (4.1)$$

where g indicates the first treatment period—thereby defining the treatment cohorts—and t is calendar time. The case $g = \infty$ indicates the potential outcome in the never-treated state. In other words, $Y_t(\infty)$ is the potential outcome at time t when a unit is not subjected to the intervention over the observed stretch of time. Listing potential outcomes that vary only by cohort and calendar time reflects the assumption of no reversibility with staggered entry.

We denote the group or cohort indicators by $\{D_S, D_{S+1}, \ldots, D_T, D_\infty\}$, where $D_\infty = 1$ indicates the never-treated. These dummy variables are mutually exclusive and exhaustive: $D_S + \cdots + D_T + D_\infty = 1$.

The ATTs of interest are now written as

$$\tau_{gr} = E\left[Y_r\left(g\right) - Y_r\left(\infty\right)|D_g = 1\right], r = g, \ldots, T; g = S, \ldots, T. \qquad (4.2)$$

For each treated cohort g, τ_{gr} denotes the ATT in each subsequent period r.

To identify the τ_{gr}, we extend the trick for the common timing case by writing

$$\begin{aligned}Y_t\left(g\right) - Y_t\left(\infty\right) = &\left[Y_t\left(g\right) - \frac{1}{\left(g-1\right)}\sum_{s=1}^{g-1} Y_s\left(g\right)\right]\\ &- \left[Y_t\left(\infty\right) - \frac{1}{\left(g-1\right)}\sum_{s=1}^{g-1} Y_s\left(\infty\right)\right]\\ &+ \frac{1}{\left(g-1\right)}\sum_{s=1}^{g-1}\left[Y_s\left(g\right) - Y_s\left(\infty\right)\right]. \qquad (4.3)\end{aligned}$$

As in the common timing case, we make a no anticipation assumption so that the third term can be dropped, which effectively allows us to use all available control units in each treated period. Here, we condition on the covariates so that not-yet-treated units can be used as part of the control group.

Assumption 4.1. **CNAS (Conditional No Anticipation, Staggered).** For $g \in \{S, \ldots, T\}$, $t \in \{1, \ldots, g-1\}$ and covariates $\mathbf{X}$,

$$E\left[Y_t\left(g\right)|D_g = 1, \mathbf{X}\right] = E\left[Y_t\left(\infty\right)|D_g = 1, \mathbf{X}\right]. \tag{4.4}$$

As in the common timing case, this assumption means that the "treatment" effects prior to the intervention are all zero. Because $s < g$ in the third sum, it follows that the expectation of the last term conditional on $D_g = 1$ is zero. Therefore,

$$\tau_{gr} = E\left[\dot{Y}_{rg}\left(g\right)|D_g = 1\right] - E\left[\dot{Y}_{rg}\left(\infty\right)|D_g = 1\right], \tag{4.5}$$

where $\dot{Y}_{rg}\left(g\right)$ and $\dot{Y}_{rg}\left(\infty\right)$ are defined as the first and second terms in (4.3), respectively. Note that the first subscript on $\dot{Y}_{rg}\left(g\right)$ and $\dot{Y}_{rg}\left(\infty\right)$ means that we are averaging all periods just prior to g and subtracting from the outcome in the current calendar time period r.

As before, the first term in (4.3) is easily estimated because we observe $\dot{Y}_{rg}(g)$ when $D_g = 1$. A parallel trends assumption, stated in terms of the never-treated state, is sufficient to identify $E[\dot{Y}_{rg}(\infty) \mid D_g = 1]$. We state the assumption conditional on a set of covariates, $\mathbf{X}$, with no covariates as a special case.

Assumption 4.2. **CPTS (Conditional PT, Staggered).** For $\mathbf{D} = (D_S, \ldots, D_T)$ and $t = 1, 2, \ldots, T$,

$$E\left[Y_t(\infty) - Y_1(\infty) \mid \mathbf{D}, \mathbf{X}\right] = E[Y_t(\infty) - Y_1(\infty) \mid \mathbf{X}], \quad t = 2, \ldots, T. \tag{4.6}$$

This assumption is used in Wooldridge (2025b). Again, it is unconfoundedness of the treatment level, as given by $\mathbf{D}$, with respect to the trend in the untreated state, $Y_t(\infty) - Y_1(\infty)$. Wooldridge (2025b) used this assumption, along with linearity of conditional means, to derive an imputation estimator and showed that it was the same as a pooled OLS estimator. Here, we show how it can be used to identify the τ_{gr} very generally.

With $\dot{Y}_{rg}\left(\infty\right)$ defined above,

$$\begin{aligned} &E\left[\dot{Y}_{rg}\left(\infty\right) \mid D_g = 1, \mathbf{X}\right] \\ &= \frac{1}{\left(g-1\right)} \sum_{s=1}^{g-1} E\left[Y_r\left(\infty\right) - Y_s\left(\infty\right) \mid D_g = 1, \mathbf{X}\right] \\ &= \frac{1}{\left(g-1\right)} \sum_{s=1}^{g-1} E\left[Y_r\left(\infty\right) - Y_s\left(\infty\right) \mid D_\infty = 1, \mathbf{X}\right] \\ &= E\left[\dot{Y}_{rg}\left(\infty\right) \mid D_\infty = 1, \mathbf{X}\right] \end{aligned} \tag{4.7}$$

where the second equality follows from CPTS and the third follows by taking the expectation outside the summation. We have shown the following.

Theorem 4.1. Under Assumption 4.1, (4.5) holds. If we add Assumption 4.2, the cohort assignments, $\mathbf{D} = (D_S, \ldots, D_T)$ are unconfounded with respect to $\dot{Y}_{rg}(\infty)$ (in the conditional mean sense), $g \in \{S, \ldots, T\}$, $r \in \left\{g, \ldots, T\right\}$, conditional on $\mathbf{X}$.

Because the vector of cohort indicators is unconfounded with respect to $\dot{Y}_{rg}\left(\infty\right)$, Theorem 4.1 implies

$$E[\dot{Y}_{rg}\left(\infty\right) \mid D_\infty = 1, \mathbf{X}] = E[\dot{Y}_{rg}\left(\infty\right) \mid D_h = 1, \mathbf{X}], \quad h = S, \ldots, T. \tag{4.8}$$

We can combine this implication of CPTS with Assumption 4.1 because, at time r, cohorts $h = r+1, \ldots, T$ have yet to be treated. Therefore,

$$E[\dot{Y}_{rg}\left(\infty\right) \mid D_h = 1, \mathbf{X}] = E[\dot{Y}_{rg}\left(h\right) \mid D_h = 1, \mathbf{X}], \quad h = r+1, \ldots, T. \tag{4.9}$$

Combined, (4.8) and (4.9) mean that, in addition to the never-treated (NT) group, we can use treatment cohorts $h \in \{r+1, \ldots, T\}$ in estimating $E[\dot{Y}_{rg}(\infty) \mid D_g = 1, \mathbf{X}]$. Incidentally, this derivation shows that if we only use the NT group as the control for each (g, r) pair, then we can drop the conditioning on $\mathbf{X}$ in Assumption 4.1. Later, we discuss what can be identified in period T without an NT group under CNAS.

We have established the following. For estimating $E[\dot{Y}_{rg}(\infty) \mid D_g = 1, \mathbf{X}]$ for $r \in \{g, \ldots, T\}$, we can use cohorts $\{r+1, \ldots, T, \infty\}$ as the control group. Define the indicator for the control group as $A_{r+1} \equiv D_{r+1} + D_{r+2} + \cdots + D_T + D_\infty$. Then, within the subpopulation $D_g + A_{r+1} = 1$, D_g is unconfounded with respect to $\dot{Y}_{rg}(\infty)$, conditional on $\mathbf{X}$. Therefore, we can apply standard treatment effect estimators after transforming the observed outcome and conditioning on this subpopulation.

Naturally, we will need an overlap assumption in order to ensure identification when using methods that condition on covariates. For τ_{gr} and using all legitimate control groups under CNAS and CPTS, the overlap assumption is

Assumption 4.3. **OVLS (Overlap, Staggered Case).** For cohorts $g \in \{S, \ldots, T\}$ and time periods $r \in \left\{g, \ldots, T\right\}$,

$$P\left(D_g = 1 \mid D_g + A_{r+1} = 1, \mathbf{X} = \mathbf{x}\right) < 1 \text{ for all } \mathbf{x} \in \text{Supp}\left(\mathbf{X}\right). \tag{4.10}$$

This condition ensures that, within the subpopulation of cohort g plus the never-treated and not-yet-treated units at time r, every treated unit has a comparable control unit.

Given data on units again indexed by i, the following simple steps lead to a general analysis. Assumptions 4.1, 4.2, and 4.3 are in force.

Procedure 4.1 **(Rolling Methods, Staggered Interventions).** Step 1. For a given $g \in \{S, \ldots, T\}$ and time period $r \in \left\{g, \ldots, T\right\}$, compute

$$\dot{Y}_{irg} \equiv Y_{ir} - \frac{1}{\left(g-1\right)} \sum_{s=1}^{g-1} Y_{is} \equiv Y_{ir} - \overline{Y}_{i,pre(g)}. \tag{4.11}$$

Step 2. Choose as the control group the units with $A_{i,r+1} = 1$ (or, if desired, a subset, such as the NT group).

Step 3. Using the subset of data with $D_{ig} + A_{i,r+1} = 1$, apply standard TE methods—such as linear RA, IPW, IPWRA, matching—to the cross section

$$\left\{\left(\dot{Y}_{irg}, D_{ig}, \mathbf{X}_i\right) : i = 1, \ldots, N\right\},$$

with D_{ig} acting as the treatment indicator.

When $\mathbf{X}_i$ has high dimension, Procedure 4.1 implies that machine learning methods can be applied after obtaining the transformation in (4.11). See, for example, Belloni, Chernozhukov, and Hansen (2014) and Chernozhukov et al. (2018).

Interestingly, when $r = g$, so that τ_{gg} is the instantaneous effect of the intervention for treatment cohort g, applying linear RA to

$$\{(\dot{Y}_{igg}, D_{ig}, \mathbf{X}_i) : i = 1, \ldots, N\},$$

with all possible control units, reproduces the POLS estimator proposed by Wooldridge (2025b). When $r > g$ this is not the case, which means, under standard assumptions, the rolling approach we propose is inefficient for the dynamic effects. The tradeoff is that we are able to apply many different kinds of estimators, including doubly robust and matching estimators.

Applying matching estimators in a rolling setting has distinct advantages over the practice of pre-matching all treated units with the never-treated group and then using standard DiD methods. For one, matches can be obtained from all not-yet-treated units in a given time period, possibly resulting in better matches. Also, if we apply standard matching software, the standard errors will reflect the uncertainty generated due to matching.

4.2. Comparison with Alternative Methods

An approach that has a similar flavor to Procedure 4.1 is the "build-the-trend" (BTT) estimator in Callaway (2023), which uses all pre-treatment periods to weight pairwise differences and uses all not-yet-treated units as controls. However, the BTT estimator is a formula based on inverse probability weighting; it is not motivated as just one of many special cases that is justified by the conclusion of unconfounded treatment. The estimator is different from that in Procedure 4.1 when we use IPW; moreover, BTT does not handle heterogeneous trends, an extension of our approach that we consider in Section 5.

Procedure 4.1 can be compared with the Callaway and Sant'Anna (2021) approach with staggered interventions. Callaway and Sant'Anna (2021) suggest using a long difference of the form

$$\mathring{Y}_{irg} \equiv Y_{ir} - Y_{i,g-1} \tag{4.12}$$

and then choosing control groups from cohorts $\{r+1, \ldots, T, \infty\}$. The transformation in (4.12) ignores control periods prior to $g-1$, whereas (4.11) gives weight $\frac{1}{(g-1)}$ to all pre-treatment outcomes.

A sketch of an identification argument that mimics that in Section 4.1 for the Callaway and Sant'Anna (2021) approach is instructive. For $g \leq t \leq T$, write

$$\begin{aligned} Y_t(g) - Y_t(\infty) = \big[Y_t(g) - Y_{g-1}(g)\big] - \big[Y_t(\infty) - Y_{g-1}(\infty)\big] \\ + \big[Y_{g-1}(g) - Y_{g-1}(\infty)\big]. \end{aligned}$$

Imposing a no anticipation assumption

$$E[Y_{g-1}(g) - Y_{g-1}(\infty) \mid D_g = 1] = 0,$$

implies

$$\begin{aligned} \tau_{gt} &= E\big[Y_t(g) - Y_t(\infty) \mid D_g = 1\big] \\ &= E\big[Y_t(g) - Y_{g-1}(g) \mid D_g = 1\big] \\ &\quad - E\big[Y_t(\infty) - Y_{g-1}(\infty) \mid D_g = 1\big]. \end{aligned} \tag{4.13}$$

The first part is always identified because

$$E\big[Y_t(g) - Y_{g-1}(g) \mid D_g = 1\big] = E\big[Y_t - Y_{g-1} \mid D_g = 1\big],$$

and so we can average the "long" difference of the units in cohort g. For the second part of (4.13), we can impose a conditional parallel trends assumption

$$\begin{aligned} E\big[Y_t(\infty) - Y_{g-1}(\infty) \mid D_S, \ldots, D_T, X\big] \\ = E\big[Y_t(\infty) - Y_{g-1}(\infty) \mid X\big], \end{aligned} \tag{4.14}$$

which is the same as saying that the vector of cohort assignments, $(D_S, \ldots, D_T)$, is unconfounded with respect to $Y_t(\infty) - Y_{g-1}(\infty)$ conditional on X.

Now the argument is the same as in Section 4.1: we can apply any treatment effect estimator based on unconfoundedness, and we can use the not-yet-treated units at time t—so cohorts $h \in \{t+1, t+2, \ldots, T, \infty\}$ as the control group.

As discussed in Callaway and Sant'Anna (2021), (4.14) only requires the CPT assumption to hold starting from period $g-1$ and allows violations before $g-1$. While Marcus and Sant'Anna (2021) describe a potential bias–variance tradeoff from using more pre-treatment periods, such a tradeoff does not always exist. Wooldridge (2025b) notes that if CPT fails into the treated periods, the Callaway and Sant'Anna (2021) estimator can have more bias than the extended TWFE (ETWFE) estimator. Like the ETWFE estimator, our approach uses all pre-treatment periods, and so it can have less bias than Callaway and Sant'Anna (2021) when CPT fails. Unfortunately, we cannot know when that will happen.

Information Set Comparison. Assuming the parallel trends assumption holds, we next compare the information sets used by the rolling, CS, and Wooldridge approaches. Table 1 summarizes the pre-treatment information used by each estimator. To illustrate the comparison, we consider treatment cohorts $g \in \{4, 5, 6\}$ and focus on estimating $ATT(4, 5)$, denoted by ★. Shaded cells indicate the pre-treatment outcome observations used by each estimator when estimating the target parameter.

The regression-based approach in Wooldridge (2025b) uses all available pre-treatment information and can be the most efficient among the three under a correctly specified linear conditional mean and standard error-components assumptions. However, this efficiency comes at the cost of potential bias if the conditional mean is misspecified.

The rolling method uses less information than the Wooldridge approach because it rules out *bad comparisons*; for example, when estimating $ATT(4, 5)$, both the rolling and CS approaches exclude group 5 at $t = 5$, since it is already treated. This may sacrifice some precision relative to Wooldridge, but, like CS, it accommodates doubly robust estimation. At the same time, compared with CS, the rolling method uses more pre-treatment information through the demeaning transformation, which can improve efficiency.

In this respect, our rolling estimator can be viewed as a middle ground between the Callaway and Sant'Anna (2021) and Wooldridge (2025b) approaches: it preserves the clean-comparison logic of CS while using more pre-treatment information, and it sacrifices some of the information used by Wooldridge in exchange for greater robustness to model misspecification.

Table 1. Pre-treatment period information used by each method.

	Callaway and Sant'Anna (2021)			Rolling method			Wooldridge (2025b)		
time	$g=4$	$g=5$	$g=6$	$g=4$	$g=5$	$g=6$	$g=4$	$g=5$	$g=6$
1									
2									
3									
4									
5	★			★			★		
6									

4.3. All Units Eventually Treated

As in Wooldridge (2023, 2025b), we can handle situations where all units are treated by $t = T$ by modifying the parallel trends assumptions and changing the specifics of the estimation. Rather than stating the CPT assumption in terms of the NT state, $Y_t(\infty)$, it is initially stated in terms of $Y_t(T)$, the state of not being treated until the final time period. Now all of the treatment effects are, initially, defined relative to $Y_t(T)$: $E\left[Y_r(g) - Y_r(T)\,|D_g = 1\right]$, $g \in \{S, \ldots, T-1\}$, $r \in \left\{g, \ldots, T\right\}$. We can no longer estimate a treatment effect for the final treated cohort because there are no available control units.

As discussed in Wooldridge (2025b), by no anticipation it follows that for $r < T$, $E\left[Y_r(g) - Y_r(T)\,|D_g = 1\right] = E\left[Y_r(g) - Y_r(\infty)\,|D_g = 1\right]$ and so, except for the final time period, the ATTs are interpreted as when we have a never-treated group. Unlike other approaches, in the final time period we can estimate the effect of being treated in earlier periods relative to being treated in the last period. This is possible using our transformation or the CS transformation, but Callaway and Sant'Anna (2021) does not explore this possibility.

In terms of estimation, the modifications to Procedure 4.1 are straightforward. In particular, $\dot{Y}_{irg}$ is computed as in (4.11) but only for $g \in \{S, \ldots, T-1\}$. In steps (2) and (3), we drop $D_{i\infty}$ everywhere—which means still choosing as the control group for cohort g in period r those units not yet treated. Then, for each $g \in \{S, \ldots, T-1\}$ and for each period $r \in \left\{g, \ldots, T\right\}$, we apply standard treatment effect estimators, as before. When $r = T$, $D_T = 1$ acts as the only control group for all cohorts first treated prior to period T.

4.4. Violations of No Anticipation. Unbalanced Panels

The no anticipation assumption requires that, prior to the first intervention period for a given treatment cohort, the potential outcomes are the same (on average) as in the never-treated state. This assumption can fail if units know that a program or policy change is approaching prior to its being actually implemented. If the NA assumption is in doubt, one can drop one or more periods prior to the intervention time and redo the analysis as a robustness check.

As an example, suppose a cohort is first treated in $g = 5$. In Procedure 4.1, one would average over periods $\{1, 2, 3, 4\}$ in obtaining the average to remove for Y_{i5} (or, one would remove a unit-specific linear trend, as in Procedure 5.1). Instead, one might drop period four altogether, or maybe even periods three and four. Any precise recommendation is context specific. It is very easy to apply any of the procedures we have recommended to cases where time periods are skipped.

Another issue that often arises in practice is unbalanced panels. With time-constant controls, unbalancedness would typically arise because of missing data on Y_{it}, possibly due to attrition. If data are missing on Y_{it} for some time periods for unit i, the demeaning or detrending is simply applied to the observed data. The mechanics of the procedure are then exactly the same. For treatment cohort g in period r, the transformed outcome ($\dot{Y}_{irg}$ or $\ddot{Y}_{irg}$) can only be used if there are enough observed data in the periods $t < g$ to compute an average (one period) or a linear trend (two periods). Of course, Y_{ir} must also be observed.

It is natural to wonder when ignoring the reason the panel is unbalanced does not cause systematic bias. Because our method removes unit-specific averages in Procedure 4.1, selection is allowed to depend on unobserved time-constant heterogeneity—just like with the usual fixed effects estimator. Selection cannot be systematically related to the shocks to $Y_{it}(\infty)$—again, just as with the FE estimator.

5. Heterogeneous Trends

One way to test for violation of the PT assumption is to estimate placebo treatment effects prior to the intervention. Callaway and Sant'Anna (2021) take this approach using their differencing method. Here, we can apply Procedure 3.1 or 4.1 to pre-treatment periods and test for effects prior to the intervention. For cohort g, it makes sense to split pre-treatment periods, $\left\{1, 2, \ldots, g-1\right\}$, into roughly equal sizes. Then, the never-treated group, or any of the groups not yet treated in $\left\{1, \ldots, g-1\right\}$ can be used as the controls. Under the null hypothesis of (conditional) PT, the tests should not find a "treatment" effect.

As motivation for adjusting Procedure 4.1 (with common timing being a special case) to allow for heterogeneous trends, express the conditional PT assumption as

$$E\left[Y_t(\infty)|\mathbf{D}, \mathbf{X}\right] = q_\infty(\mathbf{X}) + \sum_{g=S}^{T} D_g q_g(\mathbf{X}) + m_t(\mathbf{X}), \quad t = 1, \ldots, T, \tag{5.1}$$

where $q_g(\cdot)$ and $m_t(\cdot)$ are functions of the covariates, with the first not changing across time and the second not depending on the treatment cohort. As a normalization, $m_1(\mathbf{x}) \equiv 0$ for all $\mathbf{x} \in \text{Supp}(\mathbf{X})$. It is easily seen that

$$E\left[Y_t(\infty) - Y_1(\infty)|\mathbf{D}, \mathbf{X}\right] = m_t(\mathbf{X}), \quad t = 2, \ldots, T,$$

which does not depend on $\mathbf{D}$. Moreover, for $r \geq g$, the definition of $\dot{Y}_{rg}(\infty)$ implies that

$$E\left[\dot{Y}_{rg}(\infty)\,|D_\infty = 1, \mathbf{X}\right]$$
$$= m_r(\mathbf{X}) - \frac{1}{(g-1)}\sum_{s=1}^{g-1} m_s(\mathbf{X}) \equiv \dot{m}_{rg}(\mathbf{X}) \qquad (5.2)$$

and so $\dot{m}_{rg}(\cdot)$ is the conditional mean function implicit in the methods from Section 4 that use a conditional mean specification (RA or IPWRA). Because $\dot{m}_{rg}(\cdot)$ can take on positive and negative values, regression-based methods essentially assume that $\dot{m}_{rg}(\cdot)$ can be approximated by a function linear in parameters.

The representation in (5.1) suggests a way to relax parallel trends for cohorts where we have at least two pre-treatment time periods. We replace (5.1) with the following assumption.

Assumption 5.1. **CHT (Conditional Heterogeneous Trends).** For $\mathbf{D} = (D_S, \ldots, D_T)$ and $t = 1, 2, \ldots, T$,

$$E\left[Y_t(\infty)|\mathbf{D}, \mathbf{X}\right] = \eta_S(D_S \cdot t) + \cdots + \eta_T(D_T \cdot t) + q_\infty(\mathbf{X})$$
$$+ \sum_{g=S}^{T} D_g q_g(\mathbf{X}) + m_t(\mathbf{X}). \qquad (5.3)$$

Assumption 5.1 allows a separate linear trend in the never-treated state for each treatment cohort. It is easy to see that

$$E\left[Y_t(\infty) - Y_{t-1}(\infty)|\mathbf{D}, \mathbf{X}\right]$$
$$= \eta_g D_g + \cdots + \eta_T D_T + \left[m_t(\mathbf{X}) - m_{t-1}(\mathbf{X})\right] \qquad (5.4)$$

and so PT, even conditional on $\mathbf{X}$, fails. Because the trend in the never-treated state is systematically related to cohort, the estimation approaches in Sections 3 and 4 are no longer consistent.

Instead, we can linearly detrend the outcome, unit by unit, to remove the relationship between $Y_t(\infty)$ and cohort assignment. For any i, we can write

$$Y_{it}(\infty) = \mathbf{h}(\mathbf{D}_i, \mathbf{X}_i) + \mathbf{D}_i \cdot t \cdot \eta + m_t(\mathbf{X}_i) + U_{it}(\infty)$$
$$E\left[U_{it}(\infty)\,|\mathbf{D}_i, \mathbf{X}_i\right] = 0, \qquad (5.5)$$

where $\mathbf{h}(\mathbf{D}_i, \mathbf{X}_i)$ does not vary across t. For any $t \geq 2$, we can eliminate both $\mathbf{h}(\mathbf{D}_i, \mathbf{X}_i)$ and $\mathbf{D}_i \cdot t \cdot \eta$ using unit-specific linear detrending.

Equation (5.5) is an example of a heterogeneous (or random) trend model of the kind discussed in Wooldridge (2010, Section 11.7.2). Online Appendix B provides further details on the procedure for removing unit-specific linear trends by regressing the outcome on a linear function of time over the pre-treatment periods. The residuals from this regression define the detrended outcome variable, denoted as $\ddot{Y}_{irg}(\infty)$. We then show that the treatment cohort indicators, D_{ig}, are unconfounded with respect to these detrended variables, conditional on $\mathbf{X}_i$. Moreover, by Assumption 4.1, cohorts not yet treated by period r—those with $h > r$—can be used as the control group. The argument then follows the same logic as in Section 4: under Assumptions 4.1, 5.1, and 4.3, we have justified the following procedure as yielding consistent estimates of τ_{gr}.

Procedure 5.1 **(Staggered Entry).** Step 1. For a specified cohort $g \in \{S, \ldots, T\}$, run the unit-specific regressions

$$Y_{it} \text{ on } 1, t, \quad t = 1, \ldots, g-1 \qquad (5.6)$$

For $r \in \{g, \ldots, T\}$, compute the out-of-sample predictions $\hat{Y}_{irg}$ and the residuals (detrended variables) $\ddot{Y}_{irg} \equiv Y_{ir} - \hat{Y}_{irg}$. (One need not do this for units treated prior to period g.)

Step 2. Same as Procedure 4.1.

Step 3. Identical to Procedure 4.1 with $\ddot{Y}_{irg}$ replacing $\dot{Y}_{irg}$.

Procedure 5.1 is very easy to implement, only requiring many unit-specific simple regressions on a constant and time trend. The common timing case is especially easy because the regression in (5.6) is done with $g = S$ only and then the detrended outcomes $\ddot{Y}_{ir}$ are used in standard treatment effect estimation for $r = S, \ldots, T$.

In the simplest case where Procedure 5.1 can be applied, with $T = 3$ and intervention at $S = 3$ (without covariates), the resulting estimator of τ_3 is easily obtained after simple algebra:

$$\left[\left(\overline{Y}_{13} - \overline{Y}_{12}\right) - \left(\overline{Y}_{03} - \overline{Y}_{02}\right)\right] - \left[\left(\overline{Y}_{12} - \overline{Y}_{11}\right) - \left(\overline{Y}_{02} - \overline{Y}_{01}\right)\right], \qquad (5.7)$$

where the first subscript on the average is one for treated and zero for control, and the second subscript indicates time period. The first term in brackets is the usual two-period DiD estimator if the first time period is ignored. The second term is an estimate of the difference in trends prior to the intervention—often interpreted as estimating a placebo effect. The estimator in (5.7) is an example of a difference-in-difference-in-differences estimator. Procedure 5.1 allows one to control for covariates when removing an estimated pre-intervention difference in trends is not sufficient to uncover a causal effect.

Removing unit-specific trends provides additional resilience to unbalanced panels, as we are now allowing for two sources of heterogeneity—level and trend—to be correlated with selection into the sample.

A final point worth mentioning is that the unit-specific linear regressions in (5.6) do not create an incidental parameters problem when the number of time periods is small. These regressions do not require us to obtain "good" estimates of the unit-specific trends; we only need to remove the unit-specific heterogeneity specified in Assumption 5.1. The logic is the same as when we use all time periods to remove unit-specific trends, as in Wooldridge (2010, Section 11.7.2).

6. Application

In this section, we revisit the literature examining the effects of Walmart's entry on local labor markets (Basker 2005; Neumark, Zhang, and Ciccarella 2008; Brown and Butts 2025). Prior studies highlight concerns about potential violations of the parallel trends assumption. For example, Brown and Butts (2025) provide visual evidence suggesting that much of the estimated effect could be due to pre-existing trends rather than the impact of Walmart's entry (Rambachan and Roth 2023). If Walmart's entry is driven by county-level trends in economic fundamentals, observed differences in employment may reflect these trends rather than the causal effect of the store openings.

Table 2. Descriptive statistics of variables.

Variable	Obs	Mean	Std.	Min	Max
log(Retail Employment)	29,371	7.76	1.28	4.58	12.92
Share of Population above Poverty Line	29,371	0.85	0.06	0.52	0.96
Share of Population in Manufacturing	29,371	0.10	0.05	0.01	0.29
Share of Population with a High School Diploma	29,371	0.09	0.03	0.02	0.19
Treated Cohort				1986	1999
Counties	1,277				

6.1. Data

As shown in Section 5, when violations of parallel trends are driven by unit-specific linear trends, our rolling method can produce stable estimates by explicitly removing these trends. We evaluate this feature using the dataset constructed by Brown and Butts (2025), who follow Basker (2005) in using County Business Patterns data from 1964 and 1977–1999.

Following Brown and Butts (2025), the sample is restricted to counties with total employment greater than $1,500$ in 1964 and non-negative aggregate employment growth between 1964 and 1977. The Walmart store opening indicator is derived from the geographic dataset of openings in Arcidiacono et al. (2020). Counties whose first Walmart opened before 1986 are excluded. This yields a balanced panel of $1,277$ counties over 23 years, with each county observed for at least nine pre-treatment years and up to fourteen post-treatment years.

Table 2 reports descriptive statistics for the variables used in our analysis. Treatment cohorts are defined by the year of the first Walmart opening, and the outcome is county-level retail employment. The covariates are the 1980 population shares employed in manufacturing, above the poverty line, and with a high school diploma.

6.2. Estimation Results

For event time (relative period) $r = t - g$, define $WATT(r)$ as the cohort-size-weighted average of the identified cohort–time ATTs:

$$WATT(r) = \sum_{g \in \mathcal{G}_r} w(g,r)\, ATT(g,\, g+r),$$

where $\mathcal{G}_r$ is the set of cohorts for which $ATT(g, g+r)$ is identified, $w(g,r) = N_g/N_{\mathcal{G}_r}$, and $N_{\mathcal{G}_r} = \sum_{g\in\mathcal{G}_r} N_g$. The set of estimable event times depends on the outcome transformation: $r = -1$ is excluded under demeaning, while $r = -2$ and $r = -1$ are excluded under detrending, because fitting a unit-specific mean and a unit-specific linear trend requires at least one and two pre-treatment periods, respectively.

For inference on $WATT(r)$ and the event-study path $\{WATT(r) : r \in \mathcal{R}\}$, we rely on the influence-function representations implied by the stacked estimating equations. We then use these representations to implement a computationally efficient multiplier bootstrap for standard errors and simultaneous confidence bands. The procedure is implemented in the user-written Stata command `lwdid`. Further details on estimation and inference are provided in Appendices D and E of the Online Appendix, respectively. In particular, Sections E.2–E.4 derive the influence-function representations for the rolling RA, rolling IPW, and rolling IPWRA estimators.

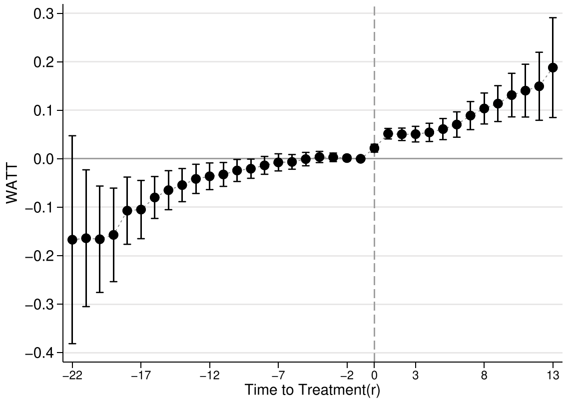
Figure 1 reports estimates of $WATT(r)$ for the effect of Walmart entry on county-level log retail employment. We present estimates from the rolling IPWRA estimator using unit-specific demeaning and unit-specific detrending, along with estimates from the doubly robust estimator of Callaway and Sant'Anna (2021). The figure also reports 95% simultaneous confidence bands for the full event-study path.

Figure 1(a) and 1(b) use different outcome transformations and are therefore normalized differently. Callaway and Sant'Anna (2021) estimates in Figure 1(a) are normalized to the last pre-treatment period, $r = -1$. In Figure 1(b), the rolling IPWRA estimates use outcomes demeaned by each county's pre-treatment mean and therefore compare treated and comparison counties in deviations from their own pre-treatment averages. Both approaches rely on their respective parallel trends conditions for identification. However, the upward pre-treatment pattern in both panels indicates differential pre-treatment dynamics between treated and comparison counties, raising concerns about violations of the relevant parallel trends conditions and the interpretation of the post-entry estimates.

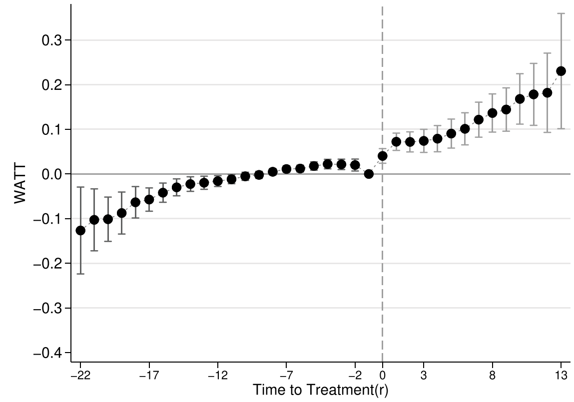
Figure 1(c) addresses this concern by using the rolling IPWRA estimator with detrending, which removes fitted county-specific linear trends based on pre-treatment observations. The resulting pre-treatment estimates are closer to zero and statistically insignificant, suggesting that the detrending procedure effectively mitigates heterogeneous pre-treatment trends across groups. The detrended specification therefore provides a more credible basis for interpreting the post-entry estimates as causal effects.

The post-treatment estimates from the detrended rolling IPWRA estimator are positive in the short run but more modest than those from Callaway and Sant'Anna (2021) or the demeaned rolling estimator. For example, $WATT(1) = 0.032$ (SE $= 0.005$), implying approximately a 3.2% increase in retail employment one year after entry, compared with about 5.2% under the CS estimator. This corresponds to about 210 additional retail jobs, which is within the 150–300 workers typically employed by a Walmart store (Basker 2005). At longer horizons, the detrended estimates return toward zero and are mostly statistically insignificant. This pattern is more consistent with modest direct employment effects than with large spillover effects.

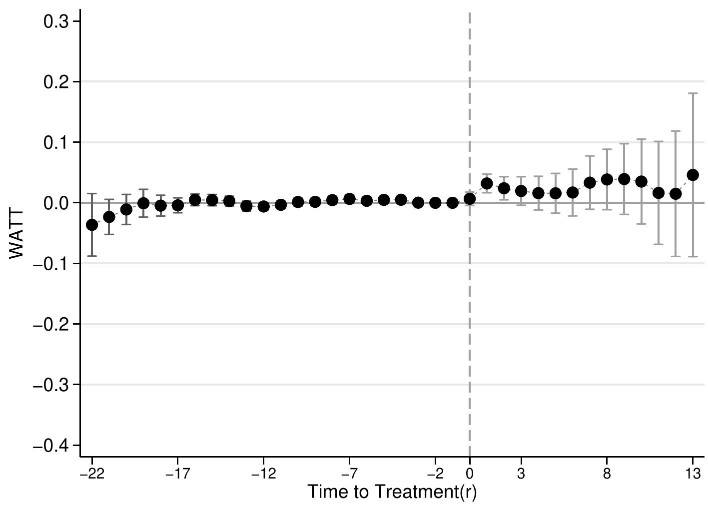
The full set of estimates is reported in Online Appendix F, and the corresponding results using wholesale employment as the outcome are reported in Online Appendix F.1. These findings highlight the empirical relevance of accounting for unit-specific trends when pre-treatment dynamics differ systematically between treated and comparison units. The results can be replicated using the user-written Stata command `lwdid`.



Panel (a) CS (2021) Approach



Panel (b) Rolling IPWRA with demeaning



Panel (c) Rolling IPWRA with detrending

Figure 1. Effects of Walmart openings on log(Retail Employment).

7. Concluding Remarks

We propose a flexible transformation approach for treatment effect estimation in panel data settings, covering both common-timing and staggered interventions. A key advantage is that, once the outcome is transformed by removing unit-specific pre-treatment means, any standard treatment effect estimator—including regression adjustment, matching, doubly robust procedures, and causal machine learning methods—can be applied to the resulting sequence of cross-sectional samples, enabling consistent estimation of heterogeneous treatment effects across cohorts and exposure lengths. The transformation also extends naturally to unit-specific detrending, which accommodates heterogeneous linear trends and thereby relaxes the parallel trends assumption even after conditioning on covariates.

Our empirical application, which examines the effects of Walmart entry on local labor markets, illustrates how detrending can be useful in settings with unit-specific heterogeneous linear trends. In such settings, widely used conditional parallel-trends methods, such as Callaway and Sant'Anna (2021), do not directly adjust for these unit-specific trend components. Together with the accompanying Stata command `lwdid`, which implements the full estimation and inference procedure, these features make the proposed rolling method a valuable tool for applied researchers.

Supplementary Materials

The online appendix includes proofs, implementation details, and additional simulation and empirical results. A single do-file provided in the online supplement replicates all results in the empirical application using the user-written Stata command lwdid (for details, see Hur, Lee, and Wooldridge 2026). The command and replication data are publicly available through Statistical Software Components (SSC). The replication data were constructed from materials provided by Brown and Butts (2025).

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Disclosure Statement

The authors report there are no competing interests to declare.

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